Khulna University of Engineering & Technology
B.Sc. Engineering 1st Year 2nd Term (Regular) Examination, 2018
Department of Materials Science and Engineering

Ph 1227

Magnetism and Nuclear Physics

Time: 3 hours | Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

Q 1. (a) What is meant by magnetic permeability and magnetic susceptibility? Draw the magnetization curve. (10)

(b) Discuss in details ferromagnetism and ferrimagnetism. (15)

(c) Find the ground state level of (i) $_{18}Ar$, (ii) $_{7}N$, (iii) $_{8}O$, (iv) $_{29}Cu$, and (v) $_{54}Xe$ (10)

Q 2. (a) Set up Clausius-Mossotti relation between polarizability and dielectric constant of a solid. Express the Clausius-Mossotti relation in terms of molar polarizability. (10)

(b) Derive the expression for the local field in a solid dielectric: (15)

$$E_{loc} = E + \frac{P}{3\varepsilon_0}$$

where the symbols have their usual meanings.

(c) Calculate Lande's g-factor for D-electron. If the total quantum number is 3, find the value of the total angular momentum quantum number. (10)

Q 3. (a) Explain space quantization in vector atom model with necessary diagram. (07)

(b) Write down all quantum numbers associated with vector atom model with their mathematical notations. (07)

(c) Write down the values of all quantum numbers for n=3. (09)

(d) Explain L-S and J-J coupling. (12)

Q 4. (a) Explain the term piezoelectricity with necessary examples. (10)

(b) State Curie-Weiss law. Derive Curie law in case of quantum theory of paramagnetism. (15)

(c) Calculate electronic polarizability of argon atom whose dielectric constant is 1.66 and density is 3.0 gm/cm^3, given that the number of argon atoms per unit volume is 1.41×10^{28} m^{-3}. (10)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

Q 5. (a) Write down the properties of nuclear force. (10)

(b) Define and draw binding energy curve and explain it with the help of fission and fusion reactions. (17)

(c) Calculate the binding energy of Deuteron. (08)

Q 6. (a) Explain nuclear fission and nuclear fusion reactions with examples. (10)

(b) Show that, for a successive radioactive disintegration, the amount of daughter substance at instant, t is given by: (15)

$$N_2 = \frac{\lambda_1 N_1^o}{\lambda_2 - \lambda_1}[e^{-\lambda_1 t} - e^{-\lambda_2 t}]$$

where the symbols have their usual meanings.

(c) One gram of Ra^{226} has an activity of nearly 1 Ci. Determine the half-life of Ra^{226}. (10)

Q 7. (a) Define radiation dosimetry. Write down different types of unit of radiation dose. (06)

(b) Discuss the syndrome of radiation. (04)

(c) Discuss the disposal of radioactive isotopes. (08)

(d) Write down the applications of radioactivity. (10)

(e) Some amount of radioactive substances of half-life 30 days is spread inside a room. Consequently, the level of radiation inside the room became 50 times the permissible level for normal occupancy of the room. After how many days the room would be safe for occupation? (07)

Q 8. (a) Distinguish between elastic and plastic materials. Explain the following terms: yield point, elastic limit, elastic fatigue, and breaking stress. (10)

(b) Prove that strain energy per unit volume is $\frac{1}{2}\times$Young's modulus$\times$(longitudinal strain)^2. (15)

(c) A copper wire with 2 meters length and 0.5 mm diameter, supports a mass of 10 kilograms. It is stretched by 2.33 mm. Calculate the Young's modulus of the wire. (10)

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B.Sc. Engineering 1st Year 2nd Term (Regular) Examination, 2018
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Math 1227
Differential Equations and Geometry

Time: 3 hours Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

Q 1. (a) Find the polar coordinates of the point (3,4,5). (10)

(b) Show that the lines whose direction–cosines are connected by $l + m + n = 0$ and $2mn + 3nl - 5lm = 0$ are perpendicular to each other. (13)

(c) Find the distance of the point P(2, 1, −3) from the line through the point A(1, 2, −3) whose direction–cosines are proportional to 1, 2, −3. (12)

Q 2. (a) Show that the equation of the plane contains the line $\frac{y}{b} + \frac{z}{c} = 1$; $x=0$ and parallel to the line $\frac{x}{a} - \frac{z}{c} = 1$; $y=0$ is $\frac{x}{a} - \frac{y}{b} - \frac{z}{c} + 1 = 0$ and if $2d$ is the shortest distance between the two lines, then prove that $d^{-2} = a^{-2} + b^{-2} + c^{-2}$ (12)

(b) If the axes are rectangular, then find the angle between the lines, $x - 2y + z = 0 = x + y - z$, $x + 2y + z = 0 = 8x + 12y + 5z$ (13)

(c) Find the distance of the point of intersection of the line $\frac{x-3}{1} = \frac{y-4}{2} = \frac{z-5}{2}$ and the plane $x + y + z = 2$ from the point (3, 4, 5). (10)

Q 3. (a) Find the magnitude and the equation of the line of the shortest distance between the two lines: $\frac{x-3}{2} = \frac{y+15}{-7} = \frac{z-9}{5}$ and $\frac{x+1}{2} = \frac{y-1}{1} = \frac{z-9}{-3}$ and also find where the line of shortest distance intersects the two lines. (15)

(b) Show that the lines $\frac{x+1}{2} = \frac{y-2}{2} = \frac{z}{1}$ and $\frac{x-1}{6} = \frac{y+1}{1} = \frac{z-3}{5}$ are coplanar. Also find their point of intersection and the equation of the plane containing them. (12)

(c) If l, m, n be the direction–cosines of any line, then prove that $l^2 + m^2 + n^2 = 1$. (08)

Q 4. (a) Find the equation of the sphere which passes through the point (α, β, γ) and the circle $z=0$, $x^2 + y^2 = a^2$ (10)

(b) Find the equation of the tangent planes to the sphere $x^2 + y^2 + z^2 + 6x - 2z + 1 = 0$ which pass through the line $3(16 - x) = 3z = 2y - 30$ (13)

(c) Consider the tetrahedron with vertices at the origin and at the points where the plane $x + 2y + 3z = 6$ intersects the coordinate axes. Compute the volume of this tetrahedron. (12)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

Q 5. (a) What is meant by the differential equation? Define the order and degree of a differential equation. Classify the differential equation with examples. Form a differential relation $xy = Ae^x + Be^{-x} + x^2$, where A and B are constants. (12)

(b) Among the following differential equations which is/are non–linear? Also give the correct reason behind the non–linearity. (10)

(i) $\frac{d^4y}{dx^4} + y^2 = \sin x$; (ii) $(\frac{d^2x}{dt^2})^2 + \omega^2 x = \sin \omega t$; (iii) $(1 - y)\frac{dy}{dx} + x \sin y = e^x$

(c) Identify and solve the following differential equation: (13)

$\left(y + \sqrt{x^2 + y^2}\right)dx - xdy = 0, y(1) = 0.$

Q 6. (a) A thermometer is removed from a room where the air temperature is 70 °F and is taken outside, where the temperature is 10 °F. After $\frac{1}{2}$ minute the thermometer reads 50 °F. What is the reading of the thermometer at t =1 min? How long will it take for the thermometer to reach 15 °F? (13)

(b) Find the particular solution of the initial value problem: (12)

$(2x - y)dx + (x + y - 3)dy = 0;\ y(0) = 2$

(c) Identify and solve the differential equation $\frac{dy}{dx} + \left(\frac{2x+1}{x}\right)y = e^{-2x}$ (10)

Q 7. (a) Find the general solution of $(D^2 - 4D + 3)y = x^2 + e^x + \cos 2x$ (12)

Where, $D \equiv \frac{d}{dx}$ and $D^2 \equiv \frac{d^2}{dx^2}$

(b) Solve by using the method of variation of parameter: $3y'' - 6y' + 6y = e^x \sec x$ (13)

(c) Find the particular solution of $y'' - y' - 12y = 0,\ y(0) = 3,\ y'(0) = 5$ (10)

Q 8. (a) Find Laplace transform of $t^2e^{-2t} + t \sin t$ (10)

(b) Find Inverse Laplace transform of $\frac{s^2+1}{s(s^2-1)(s-2)}$ (10)

(c) Using Laplace transform technique, solve the following initial value problem: (15)

$y'' + 2y' + 2y = 5\,Sin\,t,\ y(0) = y'(0) = 0$

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Hum 1227
Technical English

Time: 3 hours | Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY THREE** questions from this section in **answer script A)**

Q 1. (a) Transform the following sentences as directed: (14)

i. I am very annoyed hearing this. (Complex)
ii. One cannot be very careful about what one says. (Simple)
iii. She is one of the most regular students. (Comparative)
iv. I like him the most. (Positive)
v. We can prove that the earth is round. (Interrogative)
vi. He was so learned that he seemed to know everything. (Compound)
vii. If I were a bird! (Exclamatory)

(b) Make sentences using the following words as directed: (12)

Still (as verb); Still (as adj.), Well (as noun); Well (as adv.); Wrong (as verb); Wrong (as adv.)

(c) Change the following words as directed and make sentences with the changed words. (09)

Month (into adj.); Manager (into adj.); Light (into verb); Light (into adv.); Necessary (into verb); Necessary (into noun)

Q 2. (a) Make new words with the following suffixes and use the new words in sentence. (09)

–age, –ism, –ing, –s, –tude, –ful.

(b) Correct the following sentences: (14)

i. Dog is a pet animal.
ii. He goes to the court regularly.
iii. You what say dissatisfies people.
iv. Not only you go there but also your friend.
v. When you are hardworking, you can succeed in life.
vi. He is your true friend.
vii. The watch is out of repairs.

(c) Make sentences using the following phrases and idioms: (12)

Look blank; End in smoke; Draw the line; Eat crow; White elephant; Palmy days.

Q 3. (a) Make sentences on the following structures using the verbs given in the brackets: (14)

i. Subj. + Intransitive verb + adverbial. (Read as verb)
ii. Subj. + Linking verb + adj. complement + extension. (Become as verb)
iii. Subj. + Linking verb + noun complement + extension. (Turn as verb)
iv. Subj. + Transitive verb + Gerund as obj. (Start as verb)
v. Subj. + Transitive verb + Infinitive as obj. (Forget as verb)
vi. Subj. + Transitive verb + obj. + adj. complement. (Find as verb)
vii. Subj. + Transitive verb + obj. + noun complement. (Find as verb)

(b) Make sentences using the following modals as directed: (12)

i. May (To express guess about the future)
ii. Must (To express logical deduction in the present)
iii. Could (To express polite request)
iv. Should (To express duly in the past)
v. *Need (To express unnecessary action in the present)*
vi. Be to (To express command)

(c) Fill in the gaps of the following sentences with suitable words: (09)

i. ____he comes back, we will ____for him.
ii. Those ____read hard can ____in exam.
iii. ____he works well can ____in life.

Q 4. (a) Express the following notions/functions in sentence.

(i) Hatred, (ii) Sympathy, (iii) Separation, (iv) Honesty, (v) Wish, (vi) Determination.

(b) Complete the following sentences with subordinate clauses as directed: (1

i. ....is known to all. (noun clause)
ii. I wonder.................... (noun clause)
iii. I know the reason.................... (adjective clause)
iv. A servant....................always serves his master. (adjective clause)
v. .... , I would have helped you. (adverb clause of condition)
vi. .... , he cannot be allowed. (adverb clause of concession)
vii. Do the job.................... (adverb clause)

(c) Frame wh-questions from the underlined parts of the following answers: (09)

i. The pond is <u>10 feet deep</u>.
ii. The train departs <u>at 8 am</u>.
iii. He has been living <u>in Khulna</u> for 20 years.
iv. He has been living in Khulna <u>for 20 years</u>.
v. I borrowed <u>his bike</u>.
vi. The flowers are <u>very beautiful</u>.

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

Q 5. (a) Read the passage and answer the question that follow: (20)

Self-reliance is the pilgrim's best staff, the worker's best tool. It is the master key that unlocks all the difficulties of life. 'Help yourself and Heaven will help you' is a maxim which receives daily confirmation. Help from within always strengthens, but help from without invariably weakens the recipient. The habit of depending on others tends to weaken the intellectual faculties and paralyze the judgement. The struggle against adverse circumstances has, on the contrary, a strengthening effect, like that of the pure mountain air on an enfeebled frame. This is a lesson which is not taught in the schools nowadays. The vice of the modern system of education is that it lays down too many royal roads to knowledge. The difficulties which formerly compelled the students to think and labor for himself are now most carefully removed. The race of thorough and complete scholars is dying out. Our young men are equipped to such an extent with manuals that explain everything and guides that go everywhere, that they find no occasion for thought. Why take any trouble at all when so many are willing to relieve you of it?

<u>Questions</u>:

i. Why should we be self-reliant?
ii. What are the evil effects of dependence on others?
iii. What is the vice of modern system of education?
iv. Why is the race of thorough scholars dying out?

(b) Make a precis of the above passage (Q 5.a) with a suitable title. (15)

Q 6. (a) Write a report on the sports day you recently celebrated in your varsity. (20)

(b) Write a cause and effect paragraph on Road Accident. (15)

Q 7. (a) Write a letter to your younger brother about the importance of quality education. (20)

(b) Amplify the idea 'Uneasy lies the head that wears a crown.' (15)

Q 8. Write a composition on any one of the following topics: (35)

a) Democracy and good governance, or
b) Internet: Uses and abuses

Khulna University of Engineering & Technology

B.Sc. Engineering 1st Year 2nd Term (Regular) Examination, 2018

Department of Materials Science and Engineering

ME 1227

Engineering Mechanics

Time: 3 hours | Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.

(ii) Figure in the right margin indicates full marks.

(iii) Assume reasonable data if any is missing.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

Q 1. (a) A 5000 kg sphere rests on a smooth plane inclined at an angle $\theta=40°$ with horizontal and against a smooth vertical wall as shown in **Figure: 1(a)**. What are the reactions at the contact surfaces A and B? (17)

(b) Knowing that the tension in cable AB is 1425 N, determine the components of the force exerted on the plate at B. [see **Figure: 1(b)**] (18)

Q 2. (a) Determine the tension in each cable and the reaction at D. [see **Figure: 2(a)**] (18)

(b) In the bridge truss shown in **Figure: 2(b)**, find the forces in member AB, AF, BF and EF. (17)

Q 3. (a) Determine by direct integration the centroid of the area shown in **Figure: 3(a)**. Express your answer in terms of a and b. (18)

(b) The outside diameter of a pulley is 1 m and the cross section of its rim is as shown in **Figure: 3(b)**. Knowing that the pulley is made of steel and the density of steel is $\rho=7.8\times10^3 kg/m^3$, determine the weight of the rim. (17)

Q 4. (a) Two 8° wedges of negligible weight are used to move and position the 800 kg block as shown in **Figure: 4(a)**. Knowing that the coefficient of static friction is 0.30 at all surfaces of contact, determine the smallest force P that should be applied as shown to one of the wedges. (20)

(b) Two weights W and Q as shown in **Figure: 4(b)** are suspended from each end of a rope which passes about a stationary drum, where $\mu= {}^1/_3$. If Q is about to move downward, what is the value of ${}^Q/_W$? (15)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

Q 5. (a) Tests reveal that a normal driver takes about 0.75 seconds before he or she can react to a situation to avoid a collision. It takes about 3 seconds for a driver having 0.1% alcohol in his or her system to do the same. If such drivers are travelling on a straight road at 60 km/h and their cars can decelerate at 2 m/s^2, determine the shortest stopping distance for each from the moment they see the pedestrians. (17)

(b) The elevator shown in **Figure: 5(b)** moves downward with a constant velocity of 20 m/s. Determine (i) the velocity of the cable C, (ii) the velocity of the counterweight W, (iii) the relative velocity of the cable C with respect to the elevator. (18)

Q 6. (a) The coefficients of friction between blocks A and C and their horizontal surfaces are $\mu_s=0.24$ and $\mu_k=0.20$. Knowing that $m_A=5$ kg, $m_B=10$ kg, and $m_C=10$ kg, determine (i) the tension in the cord, (ii) the acceleration of each block. [see **Figure: 6(a)**] (17)

(b) The two dimensional motion of particle B as shown in **Figure: 6(b)** is defined by the relations $r=t^2 - {}^1/_3\, t^3$ and $\theta=2t^2$, where r is expressed in meter, t in second and θ in radians. If the particle has a mass of 2 kg and moves in a horizontal plane, determine the radial and transverse components of the force acting on the particle when (i) t = 0 and (ii) t = 1 sec. (18)

Q 7. (a) A 1.5 kg collar is attached to a spring and slides without friction along a circular rod in vertical plane as shown in **Figure: 7(a)**. The spring has an undeformed length of 150 mm and a constant k = 400 N/m. Knowing that the collar is in equilibrium at A and is given a slight push to get it moving, determine the velocity of the collar (i) as it passes through B, (ii) as it passes through C. (17)

(b) As ball A is falling shown in **Figure: 7(b)**, a juggler tosses an identical ball B which strikes ball A. The line of impact forms an angle of 30° with the vertical. Assuming the balls are frictionless and e = 0.7, determine the velocity of each ball immediately after impact. (18)

Q 8. (a) Briefly describe the different types of followers. (10)

(b) Deduce an expression for the height of a Watt governor. (08)

(c) Collar A moves upward with a constant velocity of 1.2 m/s as shown in **Figure: 8(c)**. At the instant shown when $\theta=25°$, determine (i) the angular velocity of rod AB and (ii) the velocity of collar B. (17)

Figure for Section A

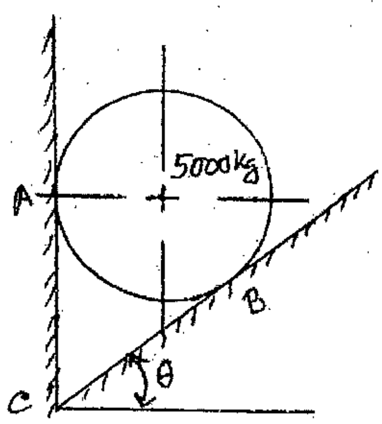

Figure: 1(a)

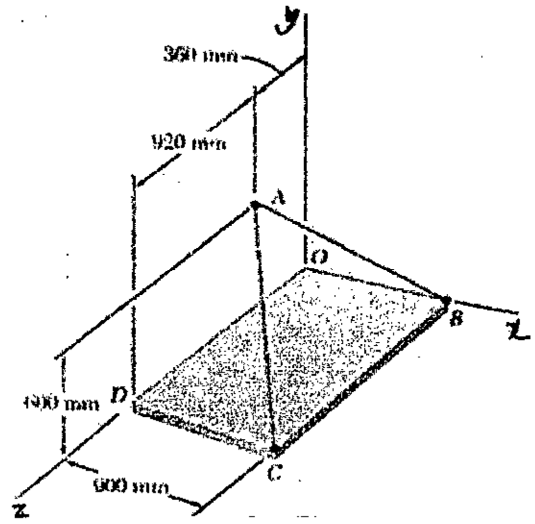

Figure: 1(b)

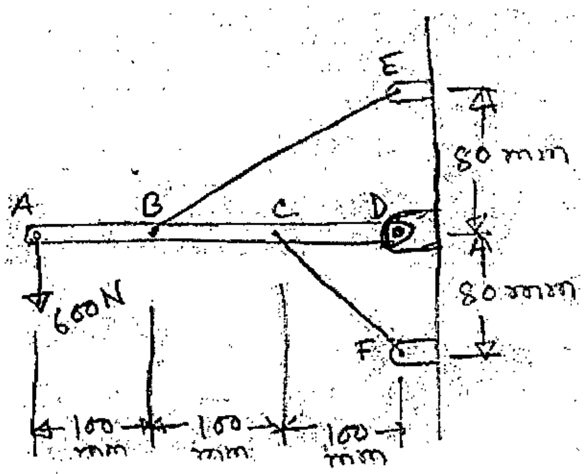

Figure: 2(a)

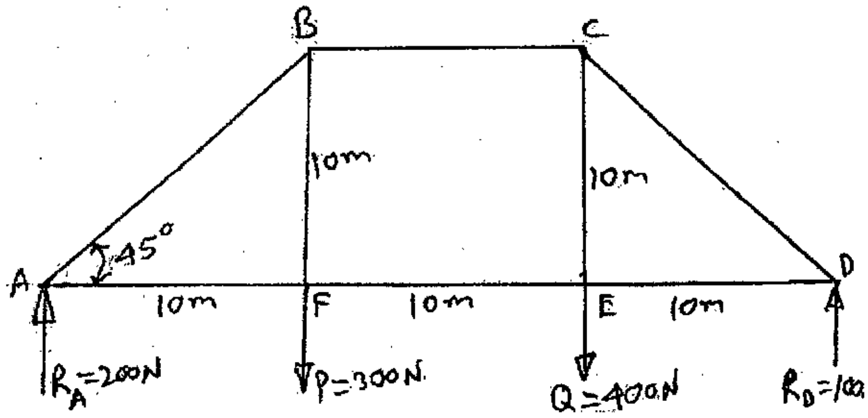

Figure: 2(b)

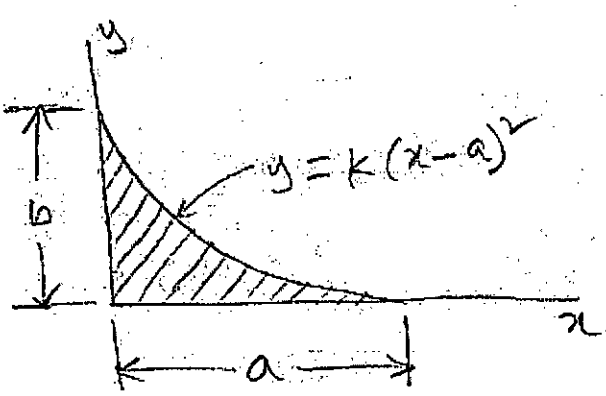

Figure: 3(a)

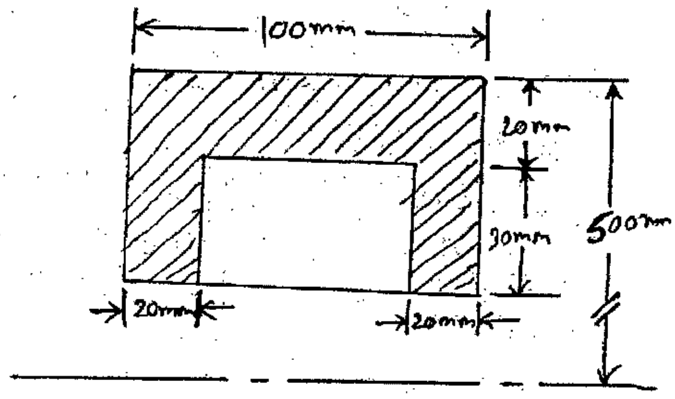

Figure: 3(b)

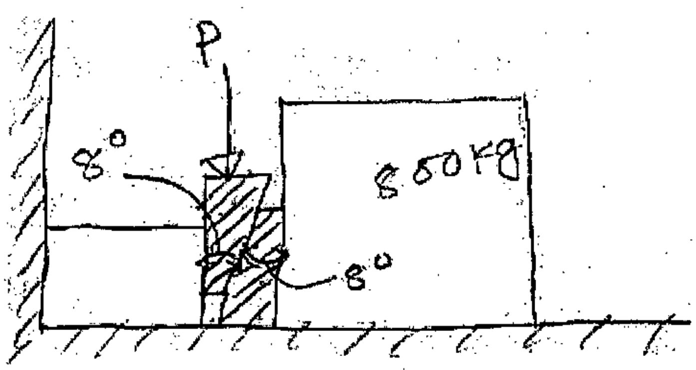

Figure: 4(a)

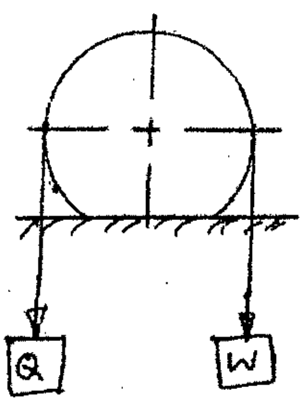

Figure: 4(b)

Figures for Section B

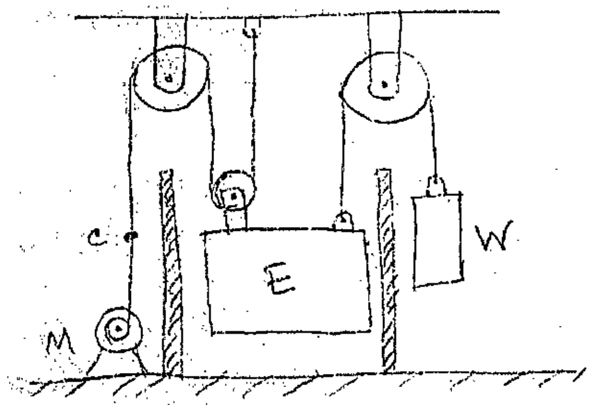

Figure: 5(b)

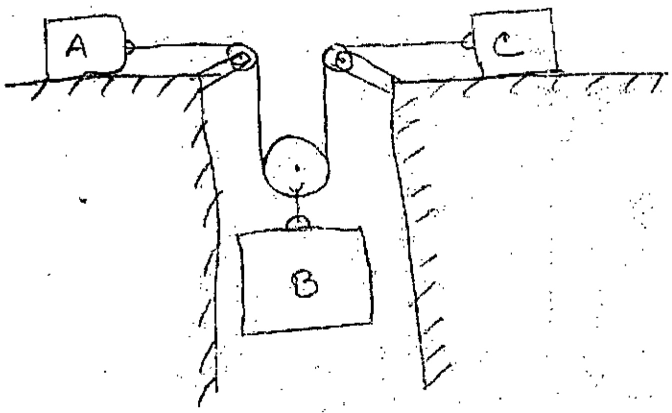

Figure: 6(a)

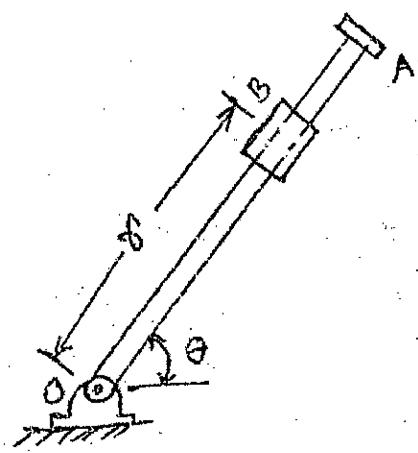

Figure: 6(b)

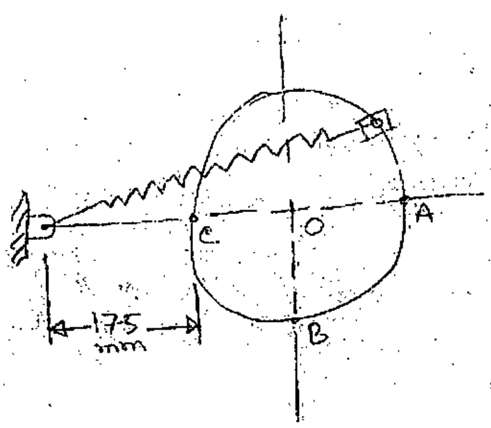

Figure: 7(a)

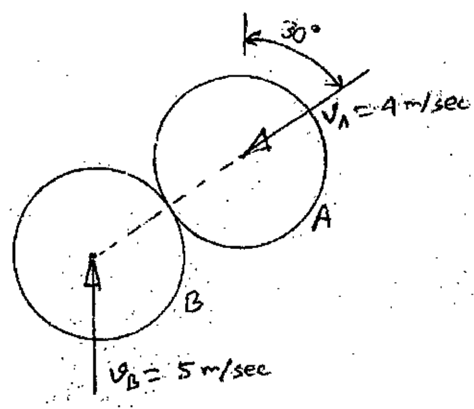

Figure: 7(b)

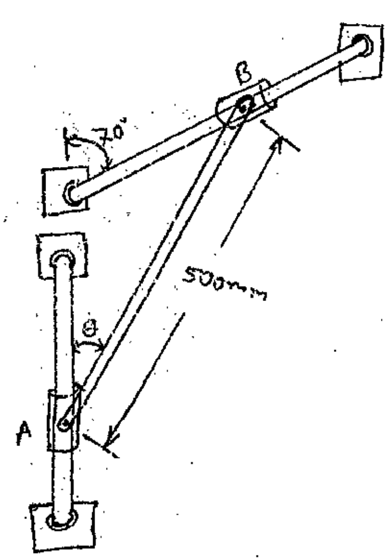

Figure: 8(c)